Appendix A: Trigonometric Identities

1	$\exp(\pm jx) = \cos x \pm j \sin x$
2	$\cos x = \frac{1}{2}\left[\exp(jx) + \exp(-jx)\right]$
3	$\sin x = \frac{1}{2j}\left[\exp(jx) - \exp(-jx)\right]$
4	$\sin^2 x + \cos^2 x = 1$
5	$\cos^2 x - \sin^2 x = \cos 2x$
6	$\cos^2 x = \frac{1}{2}\left(1 + \cos 2x\right)$
7	$\sin^2 x = \frac{1}{2}\left(1 - \cos 2x\right)$
8	$\cos^3 x = \frac{1}{4}\left(3\cos x + \cos 3x\right)$
9	$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
10	$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
11	$\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
12	$\sin x \sin y = \frac{1}{2}\left[\cos(x-y) - \cos(x+y)\right]$
13	$\cos x \cos y = \frac{1}{2}\left[\cos(x-y) + \cos(x+y)\right]$
14	$\sin x \cos y = \frac{1}{2}\left[\sin(x-y) + \sin(x+y)\right]$
15	$A\cos x + B\sin x = C\cos(x-\theta)$ where $C = \sqrt{A^2 + B^2}$ and $\theta = \tan^{-1}\dfrac{B}{A}$

Appendix B: Fourier Transform Pairs

Fourier Transform: $G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt$

Inverse Fourier Transform: $g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi f t} df$

	Time Function	**Fourier Transform**
1	$\text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1, & \lvert t \rvert \le T/2, \\ 0, & \text{otherwise.} \end{cases}$	$T \ \text{sinc}(fT)$
2	$\text{sinc}(2Wt)$	$\frac{1}{2W} \text{rect}\left(\frac{f}{2W}\right)$
3	$\exp(-at)u(t), \quad a > 0$	$\frac{1}{a + j2\pi f}$
4	$\exp(-a\lvert t \rvert), \quad a > 0$	$\frac{2a}{a^2 + (2\pi f)^2}$
5	$\exp\left[-\pi\left(\frac{t}{T}\right)^2\right]$	$T\exp\left[-\pi(fT)^2\right]$
6	$\Lambda\left(\frac{t}{T}\right) = \begin{cases} 1 - \frac{\lvert t \rvert}{T}, & \lvert t \rvert < T, \\ 0, & \text{otherwise.} \end{cases}$	$T \ \text{sinc}^2(fT)$
7	$T \ \text{sinc}^2(tT)$	$\Lambda\left(\frac{f}{T}\right)$
8	$\delta(t)$	1
9	1	$\delta(f)$
10	$\delta(t - t_0)$	$\exp(-j2\pi f t_0)$
11	$\exp(j2\pi f_c t)$	$\delta(f - f_c)$
12	$\cos(2\pi f_c t + \theta)$	$\frac{1}{2}\left[\delta(f - f_c)e^{j\theta} + \delta(f + f_c)e^{-j\theta}\right]$
13	$\sin(2\pi f_c t)$	$\frac{1}{2j}\left[\delta(f - f_c) - \delta(f + f_c)\right]$
14	$\text{sgn}(t)$	$\frac{1}{j\pi f}$
15	$\frac{1}{\pi t}$	$-j \ \text{sgn}(f)$
16	$u(t)$	$\frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$
17	$\sum_{i=-\infty}^{\infty} \delta(t - iT_0)$	$\frac{1}{T_0} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T_0}\right)$

Appendix C: Fourier Transform Properties

Property	Mathematical Description
1. Linearity	$ag_1(t) + bg_2(t) \rightleftharpoons aG_1(f) + bG_2(f)$ where a and b are constants.
2. Time Scaling	$g(at) \rightleftharpoons \frac{1}{\|a\|} G\left(\frac{f}{a}\right)$ where a is a constant.
3. Duality	If $g(t) \rightleftharpoons G(f)$, then $G(t) \rightleftharpoons g(-f)$
4. Time shifting	$g(t - t_0) \rightleftharpoons G(f)\exp(-j2\pi f t_0)$
5. Frequency shifting	$\exp(j2\pi f_c t) g(t) \rightleftharpoons G(f - f_c)$
6. Modulation Theorem	$g(t)\cos(2\pi f_c t) \rightleftharpoons \frac{1}{2}\left[G(f - f_c) + G(f + f_c)\right]$ $g(t)\sin(2\pi f_c t) \rightleftharpoons \frac{1}{2j}\left[G(f - f_c) - G(f + f_c)\right]$
7. Area under $g(t)$	$\int_{-\infty}^{\infty} g(t)\, dt = G(0)$
8. Area under $G(f)$	$g(0) = \int_{-\infty}^{\infty} G(f)\, df$
9. Differentiation in the time domain	$\frac{d}{dt} g(t) \rightleftharpoons j2\pi f G(f)$
10. Integration in the time domain	$\int_{-\infty}^{t} g(\tau)\, d\tau \rightleftharpoons \frac{1}{j2\pi f} G(f) + \frac{G(0)}{2}\delta(f)$
11. Complex conjugate functions	If $g(t) \rightleftharpoons G(f)$, then $g^*(t) \rightleftharpoons G^*(-f)$
12. Convolution in the time domain	$\int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau)\, d\tau \rightleftharpoons G_1(f) G_2(f)$
13. Multiplication in the time domain	$g_1(t) g_2(t) \rightleftharpoons \int_{-\infty}^{\infty} G_1(\lambda) G_2(f - \lambda)\, d\lambda$
14. Multiplication by t^n	$t^n g(t) \Leftrightarrow (-j2\pi)^{-1} \frac{d^n G(f)}{df^n}$